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PRISM: Positioning via RIS-Assisted Multipath Fingerprinting

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***Abstract*—Indoor Wi-Fi fingerprinting often suffers when signals observed at a single access point are low-rank, causing different locations to appear similar. We propose PRISM (Positioning via Reconfigurable Intelligent Surface Multipath Fingerprinting), an RIS-assisted localization framework that actively reshapes the wireless environment to create richer and more distinguishable fingerprints. We adopt the Bures-Wasserstein Riemannian metric, which is well suited for low-rank channel statistics, and also exploit the data geometry for improved localization. We further optimize the RIS codebook using a dual-objective Riemannian gradient method that improves covariance rank while increasing separation between nearby locations. Ray-tracing results in a realistic multi-room indoor scene show that PRISM with optimized RIS phases and PRISM-Match achieve a 0.72 m median localization error using a single antenna.**

***Index Terms*—Indoor localization, reconfigurable intelligent surface, CSI fingerprinting, Bures–Wasserstein distance, Riemannian optimization, nearest-neighbour discrimination.**

I. Introduction

INDOOR positioning with channel state information (CSI) fingerprinting is attractive precisely because it asks nothing of the existing infrastructure: record what the channel looks like at known locations, then match a new measurement to that database—no extra hardware, no ranging protocol. In a typical multi-room building, though, the channel seen at a single access point (AP) is often rank-deficient, and different rooms, corridors, or even adjacent grid points can end up with nearly indistinguishable covariance fingerprints, leaving a classifier with no reliable way to tell them apart. Even specialized large-aperture mMIMO array geometries, which have been shown to lift SINR in user-dense indoor scenarios [1], do not by themselves resolve this ambiguity: once the propagation channel itself is rank-deficient, no amount of array shaping at the AP can restore the lost diversity.

Reconfigurable Intelligent Surfaces (RIS) offer a direct way out of this [2], [3]: by programmably reshaping how signals scatter off walls and objects, an RIS can synthesize new multipath components on demand, giving each location a richer signature. Recent work has pushed on several fronts. Xu *et al.* optimize distributed RIS phase shifts via Riemannian steepest descent to minimize a closed-form localization-error probability [4]. Others focus on building a realistic fingerprint database, whether through cluster-based channel modelling [5] or hybrid propagation models benchmarked against full-wave EM simulation [6]. On the matching side, deep reinforcement learning has been used to sharpen RSSI fingerprints [7], transformer attention has been applied to CSI fingerprinting [8], and message passing has been combined with RIS phase optimization for RSS matching [9], [10]. A recent benchmark also finds k-NN fingerprinting more robust than self-organizing maps when training data is scarce [11], while other work turns to over-the-air modulation [12] or multi-floor feature selection [13] instead of RIS-based diversity. In parallel, distributed integrated sensing and communication (ISAC) systems have combined multi-target tracking with full-duplex MU-MIMO to localize emitters from existing wireless infrastructure [14], underscoring the appeal of infrastructure-light localization built on the same hardware that carries the data.

Between them, these papers cover deployment, database construction, matching, and coverage; but none fingerprint a location by its second-order channel statistics: each uses a scalar RSS value, a single snapshot, or a learned embedding, never the covariance capturing how energy spreads across antennas, frequencies, and their cross-correlations. That gap matters, since it is exactly the covariance's eigenvalue spread that determines how rank-deficient, and therefore confusable, a location's signature is. PRISM closes that gap directly: we design the RIS phase codebook for *localization discriminability* on covariance-valued fingerprints, so each fingerprint is individually rich while fingerprints at nearby grid points are pushed as far apart as possible on the manifold of positive semi-definite (PSD) matrices.

This paper's contributions are:

- A joint antenna–frequency covariance descriptor $\mathbf{R}_k \in \mathbb{C}^{D\times D}$ folding angle-of-arrival, frequency selectivity, and their cross-correlations into a single fingerprint.
- A case for the Bures–Wasserstein (BW) metric as the fingerprint similarity measure, chosen for its native handling of low-rank, noise-corrupted covariance matrices.
- A *dual-objective* RIS codebook optimizer combining a log-determinant richness term with a nearest-neighbour Frobenius discrimination term, with closed-form Wirtinger gradients.

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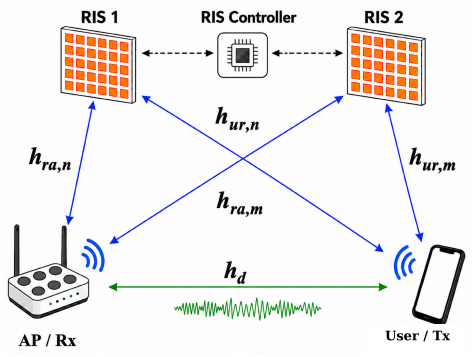


Fig. 1. System model: two RIS panels assist the uplink from User/Tx to AP/Rx.

- Validation through Sionna GPU ray tracing in a realistic multi-room indoor scene, where the optimized codebook consistently achieves the lowest mean and median positioning error of any configuration tested.

II. System Model

We consider a single-floor indoor environment with one AP, P RIS panels, and a single emitter (Tx) to localize. The AP has an N_{AP}-element ceiling array; each RIS panel has N_{elem} programmable elements, flush-mounted on an exterior wall. Localization uses the uplink, with the RIS controller setting the phases (Fig. 1). The system operates at carrier f_c with OFDM spacing Δf, selecting F subcarriers; concrete values appear in Sec. IV.

The received signal at antenna a, subcarrier f, user position m, and RIS codeword k sums the direct path, the two RIS reflections, and noise:

$$H_k(m,a,f) = H_d(m,a,f) + z(m,a,f) + \sum_{p=1}^{P}\sum_{n=1}^{N_{\mathrm{elem}}} h_{ra}^{p,a,f,n}\, W_k[p,n]\, h_{ur}^{m,p,f,n}, \quad (1)$$

where $z(m,a,f) \sim \mathcal{CN}(0,\sigma_n^2)$, h_{ra} and h_{ur} are the RIS-to-AP and user-to-RIS channels, and $W_k[p,n] = e^{j\theta_{k,p,n}}$ is the unit-modulus phase of element n on panel p under codeword k. We stack antennas and subcarriers into a joint observation,

$$\mathbf{h}_k(m) = \mathrm{vec}\big(\mathbf{H}_k(m)\big) \in \mathbb{C}^D, \qquad D = N_{\mathrm{AP}} \cdot F, \quad (2)$$

preserving the cross-antenna and cross-frequency correlations that encode both direction of arrival and delay structure in one vector.

III. PRISM Fingerprint

A. Frequency Covariance Descriptor

To build the fingerprint database, we collect L_j training snapshots around each grid point j and compress them into a single regularized sample covariance:

$$\mathbf{R}_k^{\mathrm{raw}}(j) = \frac{1}{L_j}\sum_{l=1}^{L_j} \mathbf{h}_k(l)\mathbf{h}_k(l)^H \in \mathbb{C}^{D\times D}. \quad (3)$$

The off-diagonal blocks capture $\mathbb{E}[H_k(a_1,f_1)H_k^*(a_2,f_2)]$, encoding angular and delay structure in one matrix. The K_{CB} codewords are applied *sequentially in time*: the RIS holds codeword k fixed while the L_j snapshots in (3) are collected, then switches to $k+1$, cycling through all codewords once per acquisition, with a guard interval sized to the RIS reconfiguration time. We assume the channel is static over one full cycle—reasonable since indoor coherence times exceed the switching overhead of electronically tuned RIS elements by one to two orders of magnitude—so $\mathcal{F}(j) = \{\mathbf{R}_1(j),\dots,\mathbf{R}_{K_{\mathrm{CB}}}(j)\}$ represents the same channel realization under K_{CB} scattering configurations.

Since the raw covariance in (3) can swing by orders of magnitude across the floor, we work with a trace-normalized, diagonally-loaded version,

$$\mathbf{R}_k(j) = \frac{\mathbf{R}_{0,k}(j) + \rho\,\mathbf{I}_D}{1+\rho D}, \qquad \mathbf{R}_{0,k}(j) = \frac{\mathbf{R}_k^{\mathrm{raw}}(j)}{\mathrm{tr}\big(\mathbf{R}_k^{\mathrm{raw}}(j)\big)}, \quad (4)$$

with a small regularization constant $\rho > 0$ (value in Sec. IV). Unless written as $\mathbf{R}_k^{\mathrm{raw}}$, every $\mathbf{R}_k(j)$ from here on means this normalized version.

B. Bures–Wasserstein Distance and Tangent-Space Matching

Locating a query means comparing its covariance against every database entry and returning the closest match, which raises the question of what "closest" should mean for two covariance matrices. Flattening each $\mathbf{R}_k(j)$ into a vector and using Euclidean distance ignores the geometry: covariance matrices must stay positive semi-definite, and with only $L_j \ll D$ training snapshots per grid point, every fingerprint here is already rank-deficient, so Euclidean distances get distorted the more rank-deficient they are.

We instead measure distance with the Bures–Wasserstein (BW) distance,

$$d_{\mathrm{BW}}(\mathbf{A},\mathbf{B}) = \big[\mathrm{tr}(\mathbf{A}+\mathbf{B}) - 2\,\mathrm{tr}(\mathbf{AB})^{1/2}\big]^{1/2}, \quad (5)$$

a geodesic distance on the PSD cone rooted in optimal transport theory [15]: $d_{\mathrm{BW}}(\mathbf{A},\mathbf{B})$ is the "work" needed to morph one Gaussian (covariance $\mathbf{A}$) into another (covariance $\mathbf{B}$), so it is sensitive to both shape and scale. Its exponential and logarithmic maps also remain closed-form for singular, rank-deficient covariances [16], [17], and are stable under small perturbations to near-singular matrices [18], which matters since our covariance estimates are themselves noisy.

Comparing covariances directly under (5) is too slow at inference time, so we flatten the manifold into a Euclidean tangent space at a reference point, turning nearest-neighbour search into a vector distance computation. The reference point is the BW barycenter $\overline{\mathbf{G}}_k$, computed *separately for each codeword* k via the Projection Mean Algorithm [18] over that codeword's database covariances $\{\mathbf{R}_k(j)\}_{j=1}^{K_{\mathrm{db}}}$. The BW logarithmic map at $\overline{\mathbf{G}}_k$ is the inverse of the BW exponential map [16]:

$$\log_{\overline{\mathbf{G}}_k}(\mathbf{Q}) = (\overline{\mathbf{G}}_k\mathbf{Q})^{1/2} + (\mathbf{Q}\overline{\mathbf{G}}_k)^{1/2} - 2\overline{\mathbf{G}}_k, \quad (6)$$

and we solve $\overline{\mathbf{G}}_k\mathbf{L} + \mathbf{L}\overline{\mathbf{G}}_k = \log_{\overline{\mathbf{G}}_k}(\mathbf{R}_k(j))$ for the velocity $\mathbf{L} = \mathbf{L}_k(j)$ [15], [16]. Vectorizing $\mathbf{L}$ under the plain Frobenius norm is *not* isometric to BW distance unless $\overline{\mathbf{G}}_k \propto \mathbf{I}_D$: the correct metric at $\overline{\mathbf{G}}_k$ is $g_{\overline{\mathbf{G}}_k}(\mathbf{L},\mathbf{L}) = \|\overline{\mathbf{G}}_k^{1/2}\mathbf{L}\|_F^2$ [15]. Since $\overline{\mathbf{G}}_k$ is generally anisotropic here, we embed

$$\mathbf{s}_k(j) = \mathrm{vec}\big(\overline{\mathbf{G}}_k^{1/2}\mathbf{L}_k(j)\big) \in \mathbb{C}^{D^2} \tag{7}$$

instead, so $\|\mathbf{s}_k(j) - \mathbf{s}_k(j')\|^2 = \|\overline{\mathbf{G}}_k^{1/2}(\mathbf{L}_k(j) - \mathbf{L}_k(j'))\|_F^2$ is exactly the Riemannian tangent-space distance. Location estimation is then a standard nearest-neighbour search:

$$\hat{j} = \arg\min_j \sum_{k=1}^{K_{\mathrm{CB}}} \|\mathbf{s}_k(j) - \mathbf{s}_k^q\|^2, \tag{8}$$

with $\mathbf{s}_k^q$ the query's tangent embedding. We call this matcher PRISM-Match: a one-time offline embedding followed by cheap Euclidean lookups at test time. Because the estimate is always a database grid point, accuracy is fundamentally limited by the 1 m grid spacing; we formalize this *error bound* in Sec. V.

C. Dual-Objective Formulation

The central question in codebook design for localization is: what makes a good RIS phase pattern? We argue the answer has two parts that pull in different directions. First, each individual fingerprint should be rich; its covariance matrix should have well-spread eigenvalues rather than most of its energy collapsed into a low-dimensional subspace. Second, fingerprints at nearby grid points should be discriminative; two locations a metre apart should look as different as possible to the matcher. These goals are not the same: a codebook maximizing richness alone can still produce nearly identical fingerprints across the grid. We therefore combine both into a single dual objective,

$$\mathcal{J}(\{W_k\}) = \mathcal{J}_{\mathrm{ld}}(\{W_k\}) + \lambda\,\mathcal{J}_{\mathrm{nn}}(\{W_k\}), \tag{9}$$

subject to $|W_k[p,n]| = 1$ for all k,p,n, where $\lambda \geq 0$ balances the two terms and $\mathbf{R}_k(j)$ denotes the normalized fingerprint covariance of (4).

The richness term accumulates a log-determinant penalty over all codewords and database points,

$$\mathcal{J}_{\mathrm{ld}}(\{W_k\}) = \sum_{k=1}^{K_{\mathrm{CB}}}\sum_{j=1}^{K_{\mathrm{db}}} \log\det\Big(\mathbf{I} + \tfrac{1}{\sigma_n^2}\mathbf{R}_k(j)\Big). \tag{10}$$

Since $\log\det(\cdot) = \sum_i \log\lambda_i$, any eigenvalue collapsing toward zero drives $\log\lambda_i \to -\infty$, pushing covariance energy to spread across all dimensions; the trace normalization keeps this term well-scaled regardless of absolute link power.

The discrimination term pushes apart the fingerprints of each grid point and its M spatially nearest neighbours,

$$\mathcal{J}_{\mathrm{nn}}(\{W_k\}) = \frac{1}{2}\sum_{k=1}^{K_{\mathrm{CB}}}\sum_{j=1}^{K_{\mathrm{db}}}\sum_{j'\in\mathcal{N}_M(j)} \|\mathbf{R}_k(j) - \mathbf{R}_k(j')\|_F^2, \tag{11}$$

where $\mathcal{N}_M(j)$ denotes the M nearest database neighbours of j in physical space (computed once, before optimization), and the $\frac{1}{2}$ avoids double-counting mutually linked pairs. Targeting only the nearest neighbours focuses the objective on the hardest-to-distinguish pairs.

D. Optimization via Riemannian Gradient Ascent

Both terms of (9) are differentiable in the RIS weights and share a common structure. Because $\mathbf{R}_k(j)$ is a trace-normalized, regularized function of $\mathbf{R}_k^{\mathrm{raw}}(j)$, differentiating through the normalization introduces a correction term $\Pi_{\mathbf{R}_0}(\mathbf{X}) \triangleq \mathbf{X} - \mathrm{tr}(\mathbf{X}\mathbf{R}_0)\mathbf{I}$ (the standard trace-constraint projection), which appears in both gradients below.

Theorem 1. *The Wirtinger gradient of the dual objective* (9) *with respect to* $W_k[p,n]$ *is*

$$\nabla_{W_k[p,n]}\mathcal{J} = \frac{1}{L_j}\sum_{j=1}^{K_{\mathrm{db}}}\sum_{l\in\mathcal{I}_j} \mathbf{d}_{l,p,n}^H\big(\mathbf{v}_{k,l}^{\mathrm{ld}} + \lambda\,\mathbf{v}_{k,l}^{\mathrm{nn}}\big), \tag{12}$$

where, writing $t_j = \mathrm{tr}(\mathbf{R}_k^{\mathrm{raw}}(j))$, $c = 1/(1+\rho D)$, *and* $\mathbf{A}_j = \mathbf{I} + \sigma_n^{-2}\mathbf{R}_k(j)$,

$$\mathbf{v}_{k,l}^{\mathrm{ld}} = \frac{2c}{\sigma_n^2\,t_j}\,\Pi_{\mathbf{R}_{0,k}(j)}\big(\mathbf{A}_j^{-1}\big)\,\mathbf{h}_k(l), \tag{13}$$

$$\mathbf{v}_{k,l}^{\mathrm{nn}} = \frac{2c}{t_j}\,\Pi_{\mathbf{R}_{0,k}(j)}\big(\mathbf{S}_j\big)\,\mathbf{h}_k(l), \tag{14}$$

with sensitivity matrix

$$\begin{aligned}\mathbf{S}_j = &\sum_{j'\in\mathcal{N}_M(j)}\big(\mathbf{R}_k(j) - \mathbf{R}_k(j')\big)\\ &+ \sum_{j'':\, j\in\mathcal{N}_M(j'')}\big(\mathbf{R}_k(j) - \mathbf{R}_k(j'')\big),\end{aligned} \tag{15}$$

and cascade partial derivative $\mathbf{d}_{l,p,n} = \mathrm{vec}\big(h_{ra}(p,:,:,n)\odot h_{ur}(l,p,:,:)^\top\big)$.

Proof sketch. By Jacobi's formula, the Euclidean gradients with respect to $\mathbf{R}_k(j)$ are

$$\begin{aligned}\frac{\partial\log\det(\mathbf{A}_j)}{\partial\mathbf{R}_k(j)} &= \sigma_n^{-2}\mathbf{A}_j^{-1},\\ \frac{\partial}{\partial\mathbf{R}_k(j)}\Big(\tfrac{1}{2}\|\mathbf{R}_k(j) - \mathbf{R}_k(j')\|_F^2\Big) &= \mathbf{R}_k(j) - \mathbf{R}_k(j').\end{aligned} \tag{16}$$

Summing the second term over both index roles $\mathbf{R}_k(j)$ plays in (11) gives $\mathbf{S}_j$. Pulling both gradients back through $\mathbf{R}_k^{\mathrm{raw}}(j) \to \mathbf{R}_{0,k}(j) \to \mathbf{R}_k(j)$ via the chain rule $d(\mathbf{X}/\,\mathrm{tr}(\mathbf{X}))$ produces

$$\frac{\partial}{\partial\mathbf{R}_k^{\mathrm{raw}}(j)} \;\longmapsto\; \frac{c}{t_j}\,\Pi_{\mathbf{R}_{0,k}(j)}(\cdot). \tag{17}$$

Since $\mathbf{R}_k^{\mathrm{raw}}(j) = \mathbf{H}_k(j)\mathbf{H}_k(j)^H/L_j$ is a Hermitian quadratic form in $\mathbf{H}_k(j)$, the Wirtinger cogradient rule [19]

$$\nabla_{\mathbf{H}}\,\mathrm{tr}(\mathbf{G}\mathbf{H}\mathbf{H}^H) = 2\mathbf{G}\mathbf{H} \tag{18}$$

(distinct from the bare partial derivative $\partial/\partial\mathbf{H}^* = \mathbf{G}\mathbf{H}$) combined with $\mathrm{tr}(\mathbf{M}\mathbf{u}\mathbf{v}^H) = \mathbf{v}^H\mathbf{M}\mathbf{u}$ converts (16)–(17) into the snapshot-space form (13)–(14). □

Algorithm 1 PRISM Dual-Objective RGD Codebook Optimizer

Require: $h_{ra}, h_{ur}, h_d, \sigma_n^2, \alpha, \lambda, M, N_{\text{bits}}, N_{\text{iter}}$

Initialize $W_k \leftarrow W_{\text{DFT}}$
Pre-compute $\mathcal{N}_M(j)$ from physical grid coordinates
for $i = 1$ **to** N_{iter} **do**
  Recompute $\mathbf{H}_k(l)$ via (1) for all k, l
  Form $\mathbf{R}_k^{\text{raw}}(j)$ via (3) and normalize to $\mathbf{R}_k(j)$ via (4), for all k, j
  Compute $\mathbf{S}_j$ via (15) (forward *and* reverse neighbours) for all j
  Accumulate $\mathcal{J}_{\text{ld}}$ via (10) and $\mathcal{J}_{\text{nn}}$ via (11)
  Compute combined $\mathbf{v}_{k,l}$ via (19)
  Compute $\nabla_{W_k}\mathcal{J}$ via (12)
  Project (20) and retract onto $\mathbb{T}^N$, with backtracking line search on α to guarantee $\mathcal{J}$ is non-decreasing
end for
Quantize phases to N_{bits} levels
return $\{W_k^{\text{opt}}\}$

Both gradients share the cascade partial $\mathbf{d}_{l,p,n}$ and differ only in the weighting applied to $\mathbf{h}_k(l)$:

$$\mathbf{v}_{k,l} = \mathbf{v}_{k,l}^{\text{ld}} + \lambda\, \mathbf{v}_{k,l}^{\text{nn}}. \tag{19}$$

The dual-objective gradient therefore costs a single pass over $\mathbf{d}_{l,p,n}$—no extra overhead relative to optimizing either term alone.

The unit-modulus constraint $|W_k[p,n]| = 1$ defines the complex oblique manifold $\mathbb{T}^N$. Projecting the gradient onto its tangent space [20],

$$\text{grad}_{\mathbb{T}}\mathcal{J} = \nabla - \text{Re}\{\nabla \odot W_k^*\} \odot W_k, \tag{20}$$

and retracting,

$$W_k \leftarrow \frac{W_k + \alpha\, \text{grad}_{\mathbb{T}}\mathcal{J}}{\left|W_k + \alpha\, \text{grad}_{\mathbb{T}}\mathcal{J}\right|}. \tag{21}$$

Since $\nabla_{W_k[p,n]}\mathcal{J}$ in (12) is already $2\,\partial\mathcal{J}/\partial W_k[p,n]^*$—the Brandwood ascent cogradient—it enters (20) directly, without conjugation. This tangent-projection-and-retraction step is standard for the complex circle manifold, also used in [4] for distributed RIS phase design; our gradient (Theorem 1) instead comes from the second-order covariance $\mathbf{R}_k(j)$, not a single received-signal fingerprint.

The codebook optimizer runs entirely offline on the training-split channel data and produces a fixed phase codebook that is reused for both the database and online matching, so we care more about the accuracy of the result than the cost of getting there. Each RGD iteration costs $O(K_{\text{CB}}K_{\text{db}}D^3)$, dominated by the eigendecompositions and matrix square roots in (13)–(14). With $D = 8$ this is cheap; it would only become a bottleneck if the covariance dimension grew substantially. At inference time the BW barycenter and log-map (6) are already precomputed, so an online query only costs the $O(D^2)$ nearest-neighbour search in (8).



Fig. 2. 3D room layout with AP in the center.

IV. Simulation Setup

We generate all channels with Sionna GPU ray tracing on a Mitsuba model of a 40×40 m six-room single-floor building (Fig. 2). Interior walls are 2.5 m gypsum partitions with door gaps that let signals diffract across rooms; exterior walls are concrete. The AP sits at $[0, 0, 2.5]$ m, and we flush-mount the two RIS panels on the south and east exterior walls so their angular coverage is complementary.

The link runs at Wi-Fi 2.4 GHz with 312.5 kHz OFDM spacing, keeping $F = 8$ of the 64 subcarriers from the channel-estimation phase. The AP uses a 6-element uniform circular array of radius $r = 0.06$ m, and each RIS panel carries $20 \times 20 = 400$ elements (800 total), with 6-bit phase quantization and a codebook of $K_{\text{CB}} = 4$ codewords. We restrict the covariance descriptor to a single AP antenna ($N_{\text{AP}} = 1$, $D = 8$) and leave the joint antenna–frequency case of Sec. III-B to future work.

Following the per-snapshot split of [21], we assign each of the $N = 140{,}190$ valid user positions to training or test with equal probability, giving 70,200 training and 69,990 test snapshots. The database lives on a 1 m grid with spatial-averaging radius $R_{\text{avg}} = 0.8$ m [22], yielding $K_{\text{db}} = 1402$ entries. The link operates at roughly 25 dB SNR. For the dual-objective optimizer we use a Riemannian step size $\alpha = 2$ over 100 RGD iterations, discrimination weight $\lambda = 1$, and $M = 6$ nearest neighbours per grid point. For validation we draw 1000 query locations uniformly at random, sampling each query's snapshots only from the held-out test split so no training data leaks into evaluation.

V. Results and Discussion

We evaluate four RIS configurations: no RIS, random diagonal, DFT diagonal, and our RGD-optimized codebook (PRISM); paired with four matchers: RSSI, Maximum Discriminant Projection (MDP) [23], Subspace Projection (SSP) [22], and PRISM-Match. Fig. 3 and Table I compare the matchers on the No-RIS baseline; Fig. 4 and Table II compare the four codebook configurations under PRISM-Match.

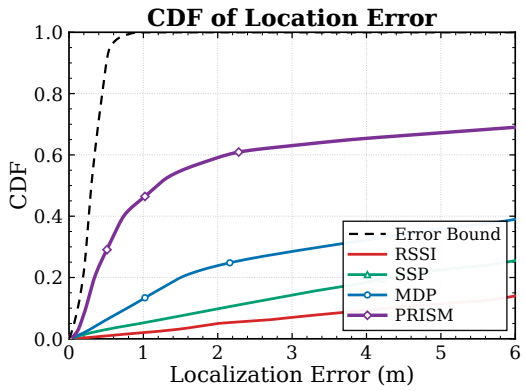


Fig. 3. CDF of localization error for the No-RIS baseline across all four matchers ($N_{\text{ant}} = 1$, SNR = 25 dB).

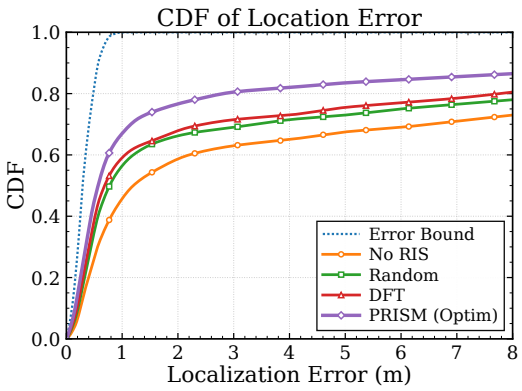


Fig. 4. PRISM-Match CDF under four codebook configurations (No RIS, Random, DFT, PRISM Optimized), $N_{\text{ant}} = 1$, SNR = 25 dB.

TABLE I
LOCALIZATION ERROR (M) ACROSS MATCHING METHODS, NO-RIS BASELINE
($N_{\text{ANT}} = 1$, $D = 8$, GRID = 1 M, $R_{\text{AVG}} = 0.8$ M, SNR = 25 DB)

Method	Mean	Med.	75th%
RSSI	18.77	18.31	26.62
SSP	17.60	16.26	27.41
MDP	14.90	10.56	25.99
PRISM-Match	**7.25**	**1.26**	**9.55**

TABLE II
PRISM-MATCH LOCALIZATION ERROR (M) ACROSS RIS CODEBOOK CONFIGURATIONS
($N_{\text{ANT}} = 1$, $D = 8$, GRID = 1 M, $R_{\text{AVG}} = 0.8$ M, SNR = 25 DB)

Codebook	Mean	Med.	75th%
No RIS	7.25	1.26	9.55
Random	6.07	0.93	6.25
DFT	5.21	0.87	4.93
Optimized (PRISM)	**3.99**	**0.72**	**1.86**

All matchers use nearest-neighbour matching: the database entry with the smallest fingerprint distance to the query is returned as the estimate, with no interpolation. Accuracy is therefore fundamentally limited by the grid resolution. We define the *error bound* at test point $\mathbf{p}$ as the Euclidean distance to its nearest grid point $\mathbf{p}^*$:

$$\text{Error Bound} = \|\mathbf{p} - \mathbf{p}^*\|_2, \tag{22}$$

which depends only on the 1 m grid geometry and is marked as the dashed curve in Fig. 4.

PRISM-Match benefits most directly from the optimized codebook. Its median error (0.72 m) beats DFT (0.87 m), Random (0.93 m), and No-RIS (1.26 m) in turn, and the gap widens in the tail: the 75th-percentile error falls from 9.55 m with no RIS to 1.86 m, an 81% reduction.

Even a random RIS already helps. With RSSI, mean error drops from 18.77 m to 16.42 m. SSP is the exception, degrading slightly under a random codebook (17.60 m $\rightarrow$ 18.38 m) because its fixed per-location projector is more sensitive to which multipath directions a given RIS draw emphasizes than it benefits from the added diversity. These three methods still cannot localize accurately in the single-antenna case, but the improvement is intuitive: even one programmable scatterer resolves some spatial ambiguity, with the largest benefit in the worst-case tail.

Moving from random or DFT to the RGD-optimized codebook brings further gains. During optimization, $\mathcal{J}_{\text{ld}}$ grows monotonically as covariances spread energy across more dimensions, while $\mathcal{J}_{\text{nn}}$ climbs alongside, pushing adjacent fingerprints apart. PRISM-Match benefits the most: its error under the optimized codebook is the lowest tested and approaches the error bound of (22). This is not coincidental – $\mathcal{J}_{\text{nn}}$ maximizes the Frobenius separation between neighbouring fingerprints, which to first order upper-bounds the BW tangent-space distance $\|\mathbf{s}_k(j) - \mathbf{s}_k(j')\|^2$ that PRISM-Match uses, so the optimizer stretches apart the same distances the matcher later relies on.

A. Robustness to SNR and Snapshot Count

To probe deployment robustness, we fix the codebook optimized at 25 dB SNR with 50 training snapshots and vary the test-time conditions.

Fig. 5 sweeps the test SNR over $\{15, 20, 25, 30\}$ dB against the No-RIS baseline. PRISM stays well ahead of No-RIS across the entire range: at 15 dB its median is only mildly degraded relative to 25 dB, and the 75th-percentile gap persists even at the low-SNR end, confirming the RIS gain is not fragile to noise.

Fig. 6 varies the online test snapshots $L_{\text{test}} \in \{5, 10, 20\}$ with the database fixed at 50 training snapshots. Performance degrades gracefully: $L_{\text{test}} = 20$ nearly matches the 30-snapshot reference, $L_{\text{test}} = 10$ loses only a modest fraction of CDF mass, and even $L_{\text{test}} = 5$ remains usable in the median, indicating viability for latency-sensitive queries.

Across all RIS scenarios, PRISM-Match remains the strongest matcher by a clear margin, with MDP a distant second, and its lead is widest under the optimized codebook – unsurprising, since that codebook was built to maximize the

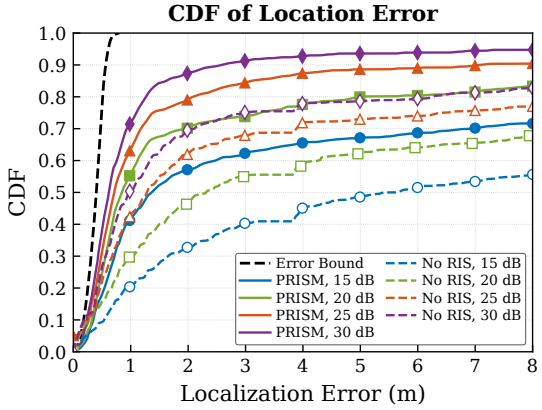


Fig. 5. PRISM-Match CDF across test SNR $\in \{15, 20, 25, 30\}$ dB vs. No-RIS; codebook trained once at 25 dB.

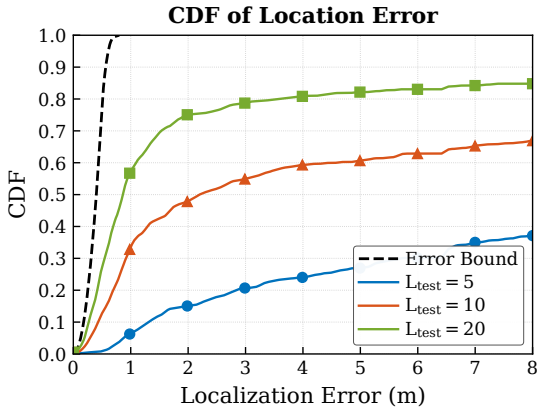


Fig. 6. PRISM-Match CDF across test-snapshot counts $L_{\text{test}} \in \{5, 10, 20\}$ at SNR = 25 dB.

distances PRISM-Match depends on. The robustness results extend this: the optimized codebook retains its advantage over a reasonable range of SNR and snapshot budgets, suiting low-cost single-antenna hardware with limited RF chains.

VI. Summary

The idea behind PRISM is simple: a RIS is a programmable scatterer we can tune to make location fingerprints more distinctive. We do that tuning with a dual-objective Riemannian gradient optimizer that spreads covariance eigenvalues via a log-determinant term and pushes apart adjacent fingerprints via a nearest-neighbour Frobenius term; the two share a cascade partial derivative, so combining them costs little more than optimizing either alone. Fingerprints are compared with the Bures–Wasserstein metric, suited to the low-rank covariances that arise under scarce training data, together with a matched tangent-space embedding (PRISM-Match) for fast search. The optimized codebook delivers the best single-antenna localization accuracy of any configuration tested.

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